

TypeScript macOS screencapture

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Introduction

, playwright . select , playwright page.screenshot ,
 . DOM .
 select . , , , IME , . play-
wright ,
 , OS . macOS screencapture , TypeScript .
E2E e2e/utils , playwright .

screencapture

. macOS .

```
# screenshot.png  
screencapture screenshot.png  
  
#  
screencapture -x screenshot.png  
  
# 3  
screencapture -T 3 screenshot.png  
  
#  
screencapture -i screenshot.png  
  
#  
screencapture -w screenshot.png  
  
#  
screencapture -t jpg screenshot.jpg
```

```
# 2
screenshot -D 2 screenshot.png
```

```
    ,
    exec
    .

, Node.js child_process, fs, path .
```

```
import {exec} from "node:child_process";
import * as fs from "node:fs";
import * as path from "node:path";
import {promisify} from "node:util";

const execAsync = promisify(exec);

/**
 *
 */
export interface ScreenCaptureOptions {
  /** */
  outputPath: string;
  /** : true */
  captureFullScreen?: boolean;
  /** : false */
  interactive?: boolean;
  /** : false */
  captureWindow?: boolean;
  /** : false */
  playSound?: boolean;
  /** : 0 */
  delay?: number;
  /** png, jpg, pdf : png */
  format?: "png" | "jpg" | "pdf" | "tiff";
  /** ID 1: , 2: */
  displayId?: number;
}
```

```
    exec
    .
```

```

export async function takeScreenshot(options: ScreenCaptureOptions): Promise<string> {
  const {
    outputPath,
    captureFullScreen = true,
    interactive = false,
    captureWindow = false,
    playSound = false,
    delay = 0,
    format = "png",
    displayId,
  } = options;

  //
  const outputDir = path.dirname(outputPath);
  if (!fs.existsSync(outputDir)) {
    fs.mkdirSync(outputDir, {recursive: true});
  }

  const args: string[] = [];

  //
  if (!playSound) {
    args.push("-x");
  }

  if (displayId !== undefined && displayId > 0) {
    args.push("-D", displayId.toString());
  }

  if (delay > 0) {
    args.push("-T", delay.toString());
  }

  args.push("-t", format);

  //
  if (interactive) {
    args.push("-i");
  } else if (captureWindow) {
    args.push("-w");
  } else if (captureFullScreen) {
    //
  }
}

```

```

args.push(outputPath);

const command = `screencapture ${args.join(" ")}`;

try {
  console.log(` : ${command}`);
  const {stderr} = await execAsync(command);

  if (stderr) {
    console.warn(` : ${stderr}`);
  }

  //
  if (!fs.existsSync(outputPath)) {
    throw new Error(` : ${outputPath}`);
  }

  console.log(` : ${outputPath}`);
  return outputPath;
} catch (error) {
  throw new Error(` : ${error}`);
}
}

```

3 .

- : screencapture . ./result/2026-09-02/ ,
- mkdirSync recursive .
- : . , playSound -x .
- : screencapture 0 , exec . existsSync .

3 . Esc , .

takeScreenshot . , .

```

/** */
export async function captureScreen(filename: string, outputDir = "./screenshots"): Promise<
  return takeScreenshot({
    outputPath: path.join(outputDir, filename),
    captureFullScreen: true,
    playSound: false,

```

```

    });
}

/**      */
export async function captureInteractive(filename: string, outputDir = "./screenshots"): Promise<string> {
    return takeScreenshot({
        outputPath: path.join(outputDir, filename),
        interactive: true,
        playSound: false,
    });
}

/**      */
export async function captureWindow(filename: string, outputDir = "./screenshots"): Promise<string> {
    return takeScreenshot({
        outputPath: path.join(outputDir, filename),
        captureWindow: true,
        playSound: false,
    });
}

/**      */
export async function captureWithDelay(
    filename: string,
    delaySeconds: number,
    outputDir = "./screenshots",
): Promise<string> {
    return takeScreenshot({
        outputPath: path.join(outputDir, filename),
        delay: delaySeconds,
        playSound: false,
    });
}

/**      */
export async function captureFromDisplay(
    filename: string,
    displayId: number,
    outputDir = "./screenshots",
): Promise<string> {
    return takeScreenshot({
        outputPath: path.join(outputDir, filename),
        displayId,
        playSound: false,
    });
}

```

```
});
}
```

```
takeScreenshot , . .
```

outputPath	string	-	
captureFullScreen	boolean	true	
interactive	boolean	false	
captureWindow	boolean	false	
playSound	boolean	false	
delay	number	0	
format	'png'	'png'	
	\\		
	'jpg'		
	\\		
	'pdf'		
	\\		
	'tiff'		
displayId	number	-	ID

```
interactive → captureWindow → captureFullScreen . true ,
.
```

```
1 .
```

```
import {captureScreen} from "./utils/screencapture";
await captureScreen("basic-screenshot.png");
```

```
takeScreenshot .
```

```
import {type ScreenCaptureOptions, takeScreenshot} from "../utils/screencapture";

const options: ScreenCaptureOptions = {
  outputPath: "../screenshots/detailed-screenshot.jpg",
  format: "jpg",
  delay: 2,
  playSound: false,
};
await takeScreenshot(options);
```

PDF , , try .

```
// PDF
await takeScreenshot({
  outputPath: "../screenshots/screenshot.pdf",
  format: "pdf",
});

// -D 2
try {
  await captureFromDisplay("display-2-screenshot.png", 2);
} catch (error) {
  console.log(" : ", error);
}
```

```
console.log(" ...");
await captureInteractive("interactive-screenshot.png");

console.log(" ...");
await captureWindow("window-screenshot.png");
```

ts-node .

```
npx ts-node utils/screencapture-examples.ts
```

- **macOS** . screencapture macOS , Linux CI . CI E2E playwright page.screenshot ,
- . IDE , .

- `.delay`
- `.headless: false`

playwright, UI, DOM, OS, macOS, playwright

E2E README AI screencapture 2